VALID: A 12-Item Validation Checklist for Financial Machine Learning

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SSRN 6508779 GitHub repo

Statistically perfect. Economically worthless.

340 strategy variants · 80 papers audited · one 12-item gate that tells them apart in ~5 minutes

The Field Today

2.5 of 12

items met

median audited paper (n = 75 empirical, 2018–2026)

Share of papers failing each dimension

No code	85%
No class balance	72%
Costs omitted	53%
No net performance	53%
BnH-only baseline	33%
Random split	7%
CPCV used	0%

This is the reporting norm, not a few bad papers.

The Strategies That Passed Everything

PASSED 9 of 340 variants

Bonferroni: 9
Holm 9 · BH-FDR 10

t up to 21.7
Harvey t > 3.0: 10 survive

SR up to 1.98
vs benchmark 0.917

Every classical correction: PASS

REALITY the same nine

PBO = 1.0
all nine — complete overfitting

AUC 0.47–0.64
near random

DSR: 0 of 340
E[max SR] under null 2.93

Corrections cannot see overfitting

Multiple-testing control is necessary — and demonstrably not sufficient.

Across all 340 variants: 52% lose money after costs · only 4.4% beat the benchmark.

All nine trade daily. CPCV rates every one fully overfit; the DSR rejects all 340.

The Gate

Stage 1 — Statistical
V1–V6 · balance, temporal split, CPCV+PBO, variance, permutation

Stage 2 — Economic
V7–V12 · net costs, cost sweep, baselines, regimes, turnover, code

Binary items · fixed order · pass/fail

False positives on signal-free data (n = 200)

AUC alone	27%	[21, 34]
+ CPCV & PBO	0%	[0, 1.9]

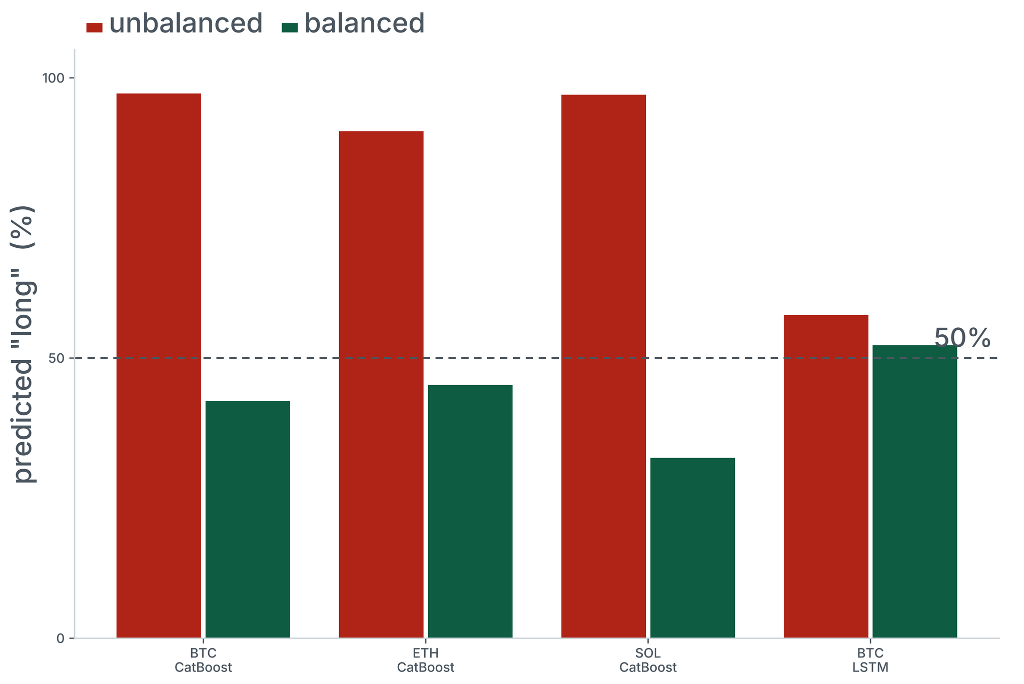
One item (V4) does it — used by 0 of 75.

Run the checklist on your own strategy
~5 minutes

orcajae.github.io/valid-framework

Bull Bias

structural class imbalance (V1–V2)



Tree models predict "long" 90–97% of the time. After balancing, AUC converges to ≈0.50 — a coin flip.

Balanced AUC (BTC / ETH / SOL)	0.50 · 0.49 · 0.51
Net SR, 1h balanced	−0.88 · −1.28 · −0.21
18 combinations, BTC mean long	85%
BTC mean balanced AUC / net SR	0.484 / −0.174

All 1h models: PBO = 1.000.

Cost Illusion

net-of-cost performance by frequency (V7–V8, V11)

Net Sharpe falls monotonically with trading frequency. At 15 minutes every single variant loses money after 18 bp costs.

	median	best	< 0
15m (n = 9)	−1.80	−1.33	100%
1h (n = 133)	−1.32	+0.32	96%
4h (n = 9)	+0.28	+0.57	44%
1d (n = 77)	+0.54	+1.98	22%

One 1h CatBoost run: gross SR 0.750 → net 0.135 at 18 bp — 82% of gross Sharpe consumed at 101 trades / yr.

340-variant corpus, 18 bp round-trip, model variants only.

Who Is This For?

adoption paths

Reviewers & PCs

Require the 12 items as a minimum reporting standard for financial ML submissions — the role TRIPOD plays in clinical prediction.

Validation & model-risk teams

A two-stage gate for incoming strategies: statistical items first, economic items second, both binary.

Allocators & operational due diligence

Twelve questions to ask before trusting a pitched Sharpe ratio.

Open source · MIT license

Coding sheet for all 80 papers and per-paper cells are public.

Score the last paper you reviewed

☐☐☐☐☐☐☐☐☐☐☐☐ median in our audit: 2.5 / 12

- ☐ **V1** Report prediction class distribution
- ☐ **V2** Test with / without class balancing
- ☐ **V3** Use temporal splitting only
- ☐ **V4** Apply CPCV with PBO
- ☐ **V5** Report Var(SR_IS)
- ☐ **V6** Include permutation tests (≥100)
- ☐ **V7** Report net performance with costs
- ☐ **V8** Cost sensitivity analysis
- ☐ **V9** Compare against simple baselines
- ☐ **V10** Evaluate in bear markets and regime transitions
- ☐ **V11** Report trade frequency
- ☐ **V12** Provide code for reproducibility

Self-audit of this paper

9 / 12 fully satisfied

Three partials disclosed: V5 (flatness criteria disagree), V8 (single 18 bp cost level), V10 (3 crypto assets only).

V1–V6 Stage 1 (statistical) V7–V12 Stage 2 (economic)

v1.0 — not a certification. The checklist has had no formal Delphi review; community revision is invited.

Selected references

Harvey, Liu & Zhu (2016). …and the Cross-Section of Expected Returns. RFS 29(1).

McLean & Pontiff (2016). Does Academic Research Destroy Return Predictability? JF 71(1).

Hou, Xue & Zhang (2020). Replicating Anomalies. RFS 33(5).

Bailey & López de Prado (2014). The Deflated Sharpe Ratio. J. Portfolio Management 40(5).

Witzany (2021). A Bayesian Approach to Measurement of Backtest Overfitting. Risks 9(1).

Kapoor & Narayanan (2023). Leakage and the Reproducibility Crisis in ML-based Science. Patterns 4(9).

Non-archival workshop paper · Full version: SSRN Working Paper 6508779

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